

HARI HARA VARAPRASAD KURRE

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Results-driven Data Engineer with 6+ years of experience building secure, scalable data platforms across healthcare, insurance, and behavioral analytics domains. Proven expertise in developing real-time and batch pipelines using **PySpark**, **Kafka**, and **dbt** across cloud ecosystems (**AWS**, **Azure**, **GCP**). Adept at modern data warehousing (**Snowflake**, **Redshift**, **BigQuery**) and orchestrating ELT workflows with **Airflow**, **Dagster**, and **Step Functions**. Skilled in enabling ML/AI pipelines via **SageMaker**, **Azure ML**, and **MLflow**, and delivering business insights through **Power BI**, **Tableau**, and **Sigma BI**. Strong focus on data quality, HIPAA compliance, reverse ETL activation (**Hightouch**), and cost-optimized cloud architecture. Certified AWS Data Engineer Associate with a track record of cross-functional collaboration and production-grade solutions.

Technical Skills:

- **Programming Languages:** Python, SQL, Scala, JavaScript, Bash, Shell
- **Big Data & Distributed Processing:** Apache Spark (PySpark, Scala), Hive, HDFS, Apache NiFi, Amazon EMR, AWS Glue, AWS Glue Crawlers, Apache Kafka, Kafka Streams, Kinesis, Kinesis Firehose, Delta Lake, Apache Beam, Apache Flink, Hadoop
- **Cloud Platforms & Services:** AWS (S3, Glue, Redshift, Lambda, EMR, Athena, DMS, SageMaker, Secrets Manager, CodePipeline, Step Functions), Azure (Data Factory, Synapse Analytics, Functions, Purview, Key Vault, Azure ML, DevOps, production support, incident management, root cause analysis, performance tuning for Azure Data Factory, Databricks, and Synapse), GCP (BigQuery, Dataproc, Dataflow, Composer, Pub/Sub), Azure Cosmos DB, Azure SQL Database, Azure Databricks Unity Catalog, REST API data integration
- **ETL/ELT & Data Integration:** dbt (Core), Talend, Sqoop, Apache NiFi, Azure Data Factory, AWS Glue, AWS DMS, Fivetran, Stitch, Airbyte, Fivetran, Custom Ingestion Frameworks with Retry Logic and State Handling, Alteryx, low-code ETL tools, ERP data integration (Oracle, SAP)
- **Data Warehousing & Databases:** Snowflake, Redshift, BigQuery, PostgreSQL, SQL Server, MongoDB, Oracle, Vertica, Cassandra, Snowflake-based data architecture, query optimization, and analytics product development
- **Streaming & Real-Time Processing:** Kafka, Kafka Streams, Apache Pulsar, Spark Structured Streaming, Kinesis, Google Pub/Sub, Redis
- **Workflow Orchestration & CI/CD:** Apache Airflow, Oozie, Jenkins, GitHub Actions, GitLab CI, Azure DevOps, AWS Step Functions, AWS CodePipeline, Prefect, dbt, Kubernetes
- **API Development & Microservices:** FastAPI, Flask, GraphQL, REST APIs, AWS API Gateway, Azure Functions, OpenAPI/Swagger
- **Infrastructure & DevOps:** Terraform, CloudFormation, Docker, Kubernetes, Jenkins, GitHub Actions, GitLab CI/CD, Prometheus, Grafana, API Gateway, GitOps Workflows, Infrastructure-as-Code (Terraform, CloudFormation)
- **Data Quality & Monitoring:** Great Expectations, Monte Carlo, Datadog, CloudWatch, PagerDuty, Prometheus, Grafana, Elasticsearch, Data Mesh Principles, Domain-Driven Design, Kimball & Data Vault Modeling, Event-Driven Architecture
- **Data Architecture & Modeling:** Lakehouse, Medallion Architecture (Bronze, Silver, Gold), Star Schema, Snowflake Schema, Data Vault, Dimensional Modeling, Microservices Architecture, Entertainment analytics data modeling, domain-driven design for media/entertainment, testing of analytics products and data sets
- **Security & Governance:** IAM (AWS), Key Vault (Azure), AWS Secrets Manager, AWS KMS, Kerberos, HIPAA-compliant data design, RBAC, Apache Atlas, Collibra, OAuth, SAML, API security protocols, API governance, PII protection, privacy-compliant data design
- **Machine Learning & AI:** TensorFlow, PyTorch, Scikit-learn, XGBoost, MLflow, Amazon SageMaker, Azure ML, Pandas, NumPy, FastAPI, Flask, Model Deployment, Model Drift Detection, Anomaly Detection, Feature Engineering, CI/CD for ML, Airflow for ML Pipeline Orchestration, Spark MLlib, AI/ML troubleshooting and optimization
- **Visualization & BI Tools:** Power BI, Tableau, Looker, Amazon QuickSight, Grafana, data storytelling, visualization best practices, Tableau dashboards, KPI tracking
- **Project & Agile Tools:** Jira, Confluence, Agile/Scrum methodologies
- **Testing & Observability:** Automated Testing in CI/CD (pytest, dbt tests), Data Observability Platforms (Monte Carlo, Datadog), Alerting Systems (PagerDuty, CloudWatch, Prometheus, Grafana)

Professional Experience:

Mayo Clinic, Rochester, Minnesota

Sep 2023 – Current

Business Analyst / Data Engineer

- Partnered with product, finance, and clinical teams to define KPIs and translate business requirements into scalable ELT pipelines using **Python**, **dbt**, and **Snowflake SQL**.
- Built reverse ETL workflows using **Hightouch** to sync curated **Snowflake** metrics to **Salesforce** and marketing platforms, improving patient outreach targeting by 30%.

- Designed event-driven pipelines using **Dagster**, and **Kafka** across **AWS** and **Azure**, delivering near-real-time insights on claims, adherence, and care coordination.
- Developed **dbt** marts for patient segmentation, readmission prediction, and cost stratification, powering dashboards in **Sigma BI**, **Power BI**, and **Tableau**. Ingested and validated healthcare data from EHR, claims, and CRM systems using **Fivetran**, **AWS Glue**, and **Azure Data Factory**, applying custom transformations and schema mapping.
- Created anomaly detection workflows using **Scikit-learn** and deployed alerts via **CloudWatch** and **PagerDuty**, reducing issue detection time by 50%.
- Led metadata governance and lineage mapping using **dbt docs**, **Snowflake** tagging, and **GitHub Actions**, improving audit readiness and traceability.
- Implemented **Great Expectations** and **dbt** tests for schema validation and business rule checks, ensuring 99%+ data accuracy in production.
- Built KPI dashboards using **Tableau**, **Power BI**, and **Sigma BI** for executive reviews and payer performance tracking.
- Automated patient churn model refresh workflows via **Dagster** and **GitHub CI/CD**, ensuring version-controlled deployments and test coverage.
- Collaborated with clinicians and data scientists on **NLP** pipelines to extract clinical entities from provider notes for care coordination.
- Migrated legacy **Excel**-based KPIs to automated **Snowflake** dashboards, reducing manual reporting by 80%. Monitored **dbt Cloud** jobs and transformation logic with **GitHub** PR checks, **Slack** alerts, and job audits.
- Supported backlog grooming and sprint planning using **Jira** and **Confluence** in an **Agile** environment.
- Built internal data dictionary tools using **Streamlit** and **Snowflake** metadata to empower non-technical users.
- Investigated and resolved data issues through root cause analysis, audit queries, and **Git** version control practices.
- Tuned **Snowflake** query performance by refactoring **CTEs**, optimizing joins, and applying clustering keys.
- Delivered internal training on **SQL** optimization, **BI** tools, and reverse **ETL** workflows for marketing and clinical teams.

Key Technologies: Snowflake, dbt, Python, SQL, Apache Airflow, Dagster, Kafka, Fivetran, AWS Glue, Azure Data Factory, Tableau, Power BI, Sigma BI, GitHub Actions, Hightouch, Scikit-learn, CloudWatch, PagerDuty, Great Expectations, Streamlit, Jira, Confluence, dbt Cloud

Tenet Healthcare, Dallas, TX

Oct 2021 – Aug 2023

Data Engineer

- **Developed, deployed, and optimized ETL pipelines** using **Python**, **AWS Glue**, **Lambda**, **Step Functions**, and **Snowflake** for behavioral health analytics, ensuring efficient data ingestion, transformation, and integration.
- Designed **event-driven, real-time data pipelines** with **Apache Airflow**, **AWS Step Functions**, and **Kafka**, enabling seamless streaming and batch data processing from healthcare provider platforms.
- **Built and optimized data models** using **SQL**, **dbt (Data Build Tool)**, and **Snowflake** applying **data modeling techniques** to reduce query complexity by 50%, improving **report generation** in **Looker**, **Tableau**, and **Mode Analytics**
- Integrated **AWS** and **GCP** for **hybrid cloud data storage**, using **Google Cloud Storage (GCS)** for backup and leveraging **AWS S3** and **Lake Formation** for structured and semi-structured data storage.
- Managed and optimized an **AWS Redshift-based data warehouse**, improving query performance by 35%, ensuring high-performance analytical workloads for clinical and business reporting.
- **Automated data quality validation** using **Great Expectations**, **AWS Glue DataBrew**, and **Python-based validation scripts**, increasing data accuracy by 40% across patient engagement and mental health assessment datasets.
- Designed **scalable, fault-tolerant data lake architecture** using **AWS S3**, **Parquet**, **Avro**, **JSON**, and **Delta Lake** formats, ensuring efficient storage and retrieval of large-scale behavioral health records.
- Implemented **data security and governance** controls using **AWS IAM**, **KMS**, **Snowflake role-based access control (RBAC)**, and row-level security, ensuring 100% compliance with **HIPAA** and **GDPR**.
- **Automated CI/CD pipeline deployments** using **Terraform**, **AWS CodePipeline**, **Docker**, and **Kubernetes (EKS)**, reducing deployment time from 3 days to 3 hours while maintaining robust **DevOps** best practices.
- Optimized batch and streaming data processing using **Apache Spark** on **AWS Glue**, reducing data processing time for large datasets by 60%.
- Designed **scalable data ingestion pipelines** integrating **Fivetran** and **Kafka**, ensuring seamless data extraction from **electronic health records (EHRs)**, clinical APIs, and third-party mental health applications.
- Developed and managed **Airflow DAGs** to schedule and orchestrate data pipeline workflows, aligning data structure with pipeline execution to optimize data movement across **AWS Glue**, **Redshift**, and **Snowflake**, supporting **task automation**.
- Collaborated with **Data Scientists** and **Machine Learning Engineers**, ensuring feature engineering pipelines efficiently transform data for AI-driven mental health treatment recommendations.
- Optimized cloud infrastructure cost and performance, leveraging **AWS Cost Explorer**, **Snowflake query optimization**, and **Redshift auto-scaling**, reducing cloud expenses by 15%.
- Worked in an **Agile** development environment, participating in daily standups, sprint planning, and retrospectives, ensuring efficient team collaboration via **Jira**, **Confluence**, and **GitHub Actions**.
- Maintained documentation and knowledge sharing through **Confluence** and **API specifications**, ensuring clarity on data schema, transformations, **business analytics** reporting and governance policies.

Key Technologies:

Python, PySpark, AWS Glue, AWS Lambda, AWS S3, AWS Redshift, AWS Step Functions, AWS IAM, AWS KMS, AWS SNS, AWS CloudWatch, AWS CodePipeline, AWS Secrets Manager, Snowflake, Google Cloud Storage (GCS), Apache Airflow, Apache Kafka, Apache Spark (AWS Glue), Fivetran, dbt, PostgreSQL, MySQL, Redshift Spectrum, Parquet, Avro, JSON, Delta Lake, Terraform, Docker, Kubernetes (EKS), Helm, Looker, Tableau, Mode Analytics, Jira, Confluence, Git, Pandas, Great Expectations, Datadog, Prefect, Jenkins

Data Engineer / Business Analyst – USAA*May 2021 – Aug 2022***San Antonio, TX**

- Partnered with **Life Company health, annuity, and insurance teams** to gather requirements, define KPIs, and design analytics solutions leveraging the **AWS cloud ecosystem**.
- Led stakeholder workshops to capture business rules for claims, underwriting, and fraud processes, translating them into technical specifications for AWS-based solutions.
- Performed **root cause analysis** on claims processing delays and data quality issues, recommending process changes that reduced cycle time by **20%**.
- Designed and deployed **ELT pipelines** using **AWS Glue, dbt, and Snowflake on AWS** as the core data warehouse, with **Databricks** leveraged for large-scale batch and advanced analytics workloads.
- Built and optimized **SQL** and **Python** transformations on **Amazon Redshift**, and Snowflake for actuarial modeling, risk analytics, and compliance dashboards.
- Integrated **Apache Kafka** with AWS-based pipelines for real-time claims and fraud analytics, improving operational responsiveness.
- Developed dashboards in **Tableau** and **Amazon QuickSight** to track claims performance, fraud detection rates, and policy lapse trends—enabling leadership to take proactive actions.
- Automated workflow orchestration using **Apache Airflow on Amazon MWAA** and **AWS Step Functions**, improving delivery timeliness by **25%**.
- Conducted **impact analysis** for new data sources and regulatory changes (**HIPAA, CMS**), ensuring alignment between compliance and technical teams.
- Deployed infrastructure and analytics workloads using **Terraform** and **AWS CloudFormation**, embedding CI/CD pipelines for automated testing and deployments.
- Managed data governance and metadata tracking with **Collibra** and **Alation** integrated with **Amazon S3** data lake storage.
- Collaborated with decision science teams to integrate predictive models into AWS production pipelines, increasing fraud detection accuracy by **15%** and reducing false positives.
- Designed AWS-based data quality frameworks that reduced regulatory reporting errors by **40%**, ensuring compliance and stakeholder trust.

Technologies: PySpark, AWS Glue, Amazon S3, Delta Lake, AWS EMR, Kafka, Spark Streaming, Kinesis Firehose, dbt, Redshift, Snowflake, Tableau, Power BI, Jenkins, GitHub Actions, Terraform, Apache Airflow, AWS Lambda, Great Expectations, AWS Glue Crawlers, AWS Glue Data Catalog, Amazon Athena, Docker, Flask, SageMaker, IAM, KMS, CloudWatch, PagerDuty, Jira, Confluence

Education: Master's in Information Technology, Webster University

Certifications: **AWS Certified Data Engineer - Associate certification**
Azure Data Engineer Associate Certification (DP-203)